SURJECTIVITY OF THE MOR MAP SUFFICES FOR THE COLIMIT IDENTITY

THE STACKS PROJECT AUTHORS

Editorial note. No omissions: the argument is fully repeated in the counted span rather than citing the characterization proposition.

Allowed standard prerequisites (editorial).

- Limit closed nonempty. Limits, Lemma lemma-limit-closed-nonempty

Source reference. Original proposition 01ZC: sources/stacks-chow/tags.txt, source lines 2587–2587. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00QO: sources/stacks-chow/tags.txt, source lines 974–974. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01Z3: sources/stacks-chow/tags.txt, source lines 2578–2578. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. If for every directed limit $T = \lim_{i \in I} T_i$ of affine schemes over S the map*

$$\operatorname{colim} \operatorname{Mor}_S(T_i, X) \longrightarrow \operatorname{Mor}_S(T, X)$$

is surjective, then f is locally of finite presentation. In other words, in Proposition 01ZC parts (2) and (3) it suffices to check surjectivity of the map.

Proof. The proof is exactly the same as the proof of the implication "(2) implies (1)" in Proposition 01ZC. Choose any affine opens $U \subset X$ and $V \subset S$ such that $f(U) \subset V$. We have to show that $\mathcal{O}_S(V) \to \mathcal{O}_X(U)$ is of finite presentation. Let $(A_i, \varphi_{ii'})$ be a directed system of $\mathcal{O}_S(V)$-algebras. Set $A = \operatorname{colim}_i A_i$. According to Algebra, Lemma 00QO it suffices to show that

$$\operatorname{colim}_i \operatorname{Hom}_{\mathcal{O}_S(V)}(\mathcal{O}_X(U), A_i) \to \operatorname{Hom}_{\mathcal{O}_S(V)}(\mathcal{O}_X(U), A)$$

is surjective. Consider the schemes $T_i = \operatorname{Spec}(A_i)$. They form an inverse system of V-schemes over I with transition morphisms $f_{ii'} : T_i \to T_{i'}$ induced by the $\mathcal{O}_S(V)$-algebra maps $\varphi_{i'i}$. Set $T := \operatorname{Spec}(A) = \lim_i T_i$. The formula above becomes in terms of morphism sets of schemes

$$\operatorname{colim}_i \operatorname{Mor}_V(T_i, U) \to \operatorname{Mor}_V(\lim_i T_i, U)$$

We first observe that $\operatorname{Mor}_V(T_i, U) = \operatorname{Mor}_S(T_i, U)$ and $\operatorname{Mor}_V(T, U) = \operatorname{Mor}_S(T, U)$. Hence we have to show that

$$\operatorname{colim}_i \operatorname{Mor}_S(T_i, U) \to \operatorname{Mor}_S(\lim_i T_i, U)$$

is surjective and we are given that

$$\operatorname{colim}_i \operatorname{Mor}_S(T_i, X) \to \operatorname{Mor}_S(\lim_i T_i, X)$$

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is surjective. Hence it suffices to prove that given a morphism $g_i : T_i \to X$ over S such that the composition $T \to T_i \to X$ ends up in U there exists some $i' \geq i$ such that the composition $g_{i'} : T_{i'} \to T_i \to X$ ends up in U. Denote $Z_{i'} = g_{i'}^{-1}(X \setminus U)$. Assume each $Z_{i'}$ is nonempty to get a contradiction. By Lemma 01Z3 there exists a point t of T which is mapped into $Z_{i'}$ for all $i' \geq i$. Such a point is not mapped into U. A contradiction. $\square$